SECTION B – API Endpoint Plan

Authentication Endpoints.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register a new user with role selection	None	{ email, password, firstName, lastName, role, (optional: organisationName for Organiser, dateOfBirth for Participant) }	201 Created - User object with ID and role 400 Bad Request - Validation errors 409 Conflict - Email already exists
POST	/api/auth/login	Authenticate user and create session	None	{ email, password }	200 OK - User object with role and session 401 Unauthorized - Invalid credentials
POST	/api/auth/logout	End user session	Any (Logged in)	None	200 OK - Logout successful 401 Unauthorized - Not logged in
GET	/api/auth/me	Get current authenticated user info	Any (Logged in)	None	200 OK - User object 401 Unauthorized - Not logged in

User Profile Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Get current user's profile with role-specific details	Any (Logged in)	None	200 OK - User profile with role-specific data 401 Unauthorized - Not logged in
PUT	/api/users/profile	Update current user's profile	Any (Logged in)	{ firstName, lastName, (role-specific fields) }	200 OK - Updated user profile 400 Bad Request - Validation errors 401 Unauthorized - Not logged in

EVENT Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Get all events with optional filters	None	Query Parameters: type, location, dateFrom, dateTo	200 OK - List of events 400 Bad Request - Invalid filter parameters
GET	/api/events/{id}	Get event by ID with categories	None	None	200 OK - Event with categories 404 Not

					Found - Event does not exist
POST	/api/events	Create a new event	Organiser	{ name, description, date, location, distance, eventType }	201 Created - New event object 400 Bad Request - Validation errors 401 Unauthorized - Not logged in 403 Forbidden - Not an Organiser
PUT	/api/events/{id}	Update event details	Organiser	{ name, description, date, location, distance, eventType }	200 OK - Updated event 400 Bad Request - Validation errors 401 Unauthorized - Not logged in 403 Forbidden - Not an Organiser or not the event owner 404 Not Found - Event does not exist
DELETE	/api/events/{id}	Delete an event	Organiser	None	204 No Content - Event deleted 401 Unauthorized - Not logged in 403 Forbidden - Not an Organiser or not the event owner 404 Not Found - Event does not exist

Event Image Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{id}/images	Upload event banner image	Organiser	Multipart form data: image file	201 Created - Image object with URL 400 Bad Request - Invalid file 401 Unauthorized - Not logged in 403 Forbidden - Not event owner 404 Not Found - Event does not exist
DELETE	/api/events/{id}/images/{imageId}	Delete event image	Organiser	None	204 No Content - Image deleted 401 Unauthorized - Not logged in 403 Forbidden - Not event owner 404 Not Found - Event or image not found

Category Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Get categories for an event	None	None	200 OK - List of categories 404 Not Found - Event does not exist
POST	/api/events/{eventId}/categories	Create a category for an even	Organiser	{ name, description }	201 Created - New category object 400 Bad Request - Validation errors 401 Unauthorized - Not logged in 403 Forbidden - Not event owner 404 Not Found - Event does not exist

PUT	/api/categories/{id}	Update category	Organiser	{ name, description }	200 OK - Updated category 400 Bad Request - Validation errors 401 Unauthorized - Not logged in 403 Forbidden - Not event owner 404 Not Found - Category does not exist
DELETE	/api/categories/{id}	Delete category	Organiser	None	204 No Content - Category deleted 401 Unauthorized - Not logged in 403 Forbidden - Not event owner 404 Not Found - Category does not exist 409 Conflict - Category has enrolments

Enrolment Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/enrolments	Get all enrolments for an event	Organiser	None	200 OK - List of enrolments with participant details 401 Unauthorized - Not logged in 403 Forbidden - Not event owner 404 Not Found - Event does not exist
POST	/api/events/{eventId}/enrol	Enrol participant in event	Participant	{ categoryId }	201 Created - Enrolment

					object 400 Bad Request - Validation errors 401 Unauthorized - Not logged in 403 Forbidden - Not a Participant 404 Not Found - Event or category not found 409 Conflict - Already enrolled
GET	/api/participants/enrolments	Get current participant's enrolments	Participant	None	200 OK - List of enrolments with event and category details 401 Unauthorized - Not logged in 403 Forbidden - Not a Participant
PUT	/api/enrolments/{id}/status	Update enrolment status	Organiser	{ status }	200 OK - Updated enrolment 400 Bad Request - Invalid status 401 Unauthorized - Not

					logged in 403 Forbidden - Not event owner 404 Not Found - Enrolment not found

Result Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/results	Capture results for participants	Organiser	[{ enrolmentId, finishTime, finishingPosition }]	201 Created - Results created 400 Bad Request - Validation errors 401

					Unauthorized - Not logged in 403 Forbidden - Not event owner 404 Not Found - Event not found
PUT	/api/results/{id}	Update a result	Organiser	{ finishTime, finishingPosition }	200 OK - Updated result 400 Bad Request - Validation errors 401 Unauthorized - Not logged in 403 Forbidden - Not event owner 404 Not Found - Result not found
GET	/api/participants/results	Get current participant's results	Participant	None	200 OK - List of results with event details 401 Unauthorized - Not logged in 403 Forbidden - Not a Participant
GET	/api/events/{eventId}/results	Get all results for an event	Organiser	None	200 OK - List of results with participant

					details 401 Unauthorized - Not logged in 403 Forbidden - Not event owner 404 Not Found - Event not found

Admin endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/organisers/dashboard	Get organiser dashboard summary	Organiser	None	200 OK - Dashboard data (events, enrolments, upcoming events) 401 Unauthorized - Not logged in 403 Forbidden - Not an Organiser
GET	/api/events/upcoming	Get upcoming events	None	Query Parameters: days (default 30)	200 OK - List of upcoming events